Imputing Total Medical Expenditure and the Medical Spending–Wealth Gradient

Amin Shirazian [*]

May 1, 2026

Preliminary and incomplete. Comments welcome.

This version: May 1, 2026

Abstract

Using out-of-pocket medical spending data from the Health and Retirement Study (HRS), this paper documents a sharp decline in medical spending among single-member households aged 55–64 in the second and third wealth deciles. Because out-of-pocket spending may understate true medical costs when households are well-insured, we impute total medical expenditure using the Medical Expenditure Panel Survey (MEPS) as a donor sample. The dip persists in total spending, indicating that these households are not heavily insured. We propose a stylized model in which a lower asset threshold for means-tested insurance generates this pattern through strategic asset decumulation.

1 Introduction

Understanding the relationship between medical spending and wealth requires a measure of total medical expenditure—not just the out-of-pocket component that households pay directly. Out-of-pocket spending is widely available in household surveys, but it captures only a fraction of total costs. Insurance absorbs the remainder, and the share absorbed varies systematically across the wealth distribution: wealthier households tend to carry more generous coverage, so out-of-pocket spending compresses the true spending gradient. How large this compression is, and whether it differs across household types, are open empirical questions that cannot be answered without a measure of total medical costs.

This paper constructs such a measure. Using the Medical Expenditure Panel Survey (MEPS), in which total

[*] Amin Shirazian, Ph.D. Candidate, Toronto Metropolitan University

medical expenditure is verified directly by healthcare providers, we estimate the relationship between out-of-pocket and total spending and use it to impute total medical expenditure in the Health and Retirement Study (HRS). The HRS contains detailed data on household wealth but records only out-of-pocket medical costs. The imputation bridges this gap, producing a dataset in which both total medical spending and wealth are observed for the same households.

We make two contributions. First, utilizing the imputation procedure by Blundell et al. (2004), we validate the imputation empirically. Within MEPS, the imputed values closely reproduce the distribution of observed total spending, with mean differences below 0.1 log points across all waves. When applied to the HRS, the imputed means diverge from the MEPS benchmark due to differences in out-of-pocket spending distributions across the two surveys. We propose a simple adjustment that removes this divergence, confirming that the imputation targets the correct population moments.

Second, we show that using the imputed variable as the dependent variable in an OLS regression preserves consistency of the slope coefficients under standard exogeneity assumptions, which we argue are credible given the provider-verified structure of the MEPS data. We use the imputed measure to document the medical spending–wealth gradient and compare it to the gradient implied by out-of-pocket spending alone. For couple households, total medical spending rises monotonically across the wealth distribution, with the wealthiest decile spending approximately 89 percent more than the poorest. The out-of-pocket gradient is roughly half as steep, reaching only 47 percent at the top decile—a gap exceeding 30 percentage points. For singles, the two measures track more closely, and both exhibit a U-shaped pattern in which the very poorest households spend more than those with near-zero wealth. These findings indicate that studies relying solely on out-of-pocket data may substantially understate the relationship between medical expenditure and wealth, particularly for couple households.

This paper relates to several strands of the literature. Skinner (1987) proposes a linear specification to impute total consumption expenditure from its components, an approach widely adopted in the consumption literature (Palumbo (1999), Ziliak (1998), Browning and Leth-Petersen (2003)). Blundell et al. (2004) extend this framework with adjustments designed to satisfy moment-consistency conditions, and we build directly on their method. A recurring concern in such imputation strategies is measurement error (Campos and Reggio (2014)), for which proposed remedies include instrumental variables (Blundell et al. (2008)), multiple measures (Browning and Crossley (2009)), and multiple proxies (Bollinger and Minier (2015)). We mitigate these concerns by relying on the provider-verified expenditure data in MEPS and by directly assessing the distributional properties of the imputed measure. Using imputed variables in regression raises additional issues: standard inference for moments can be distorted (Attanasio and Pistaferri (2014)), and using an imputed variable as either a dependent (Crossley et al. (2022)) or independent variable (Wooldridge (2010)) can bias estimates. We address the dependent-variable case formally and provide conditions under which the OLS estimator remains consistent.

The remainder of the paper is organized as follows. Section 2 develops the imputation procedure and establishes its theoretical properties. Section 3 proves consistency of OLS estimation with the imputed dependent variable. Section 4 describes the data, and Section 5 validates the imputation empirically. Section 6 presents the medical spending–wealth gradient and the comparison with out-of-pocket spending. Section 7 provides the theoretical framework to explain the main finding. Section 8 concludes.

2 Imputation Procedures

This section applies the method of Blundell et al. (2004) to impute total medical spending.[1] It then states conditions under which the imputations are consistent in mean and variance. Finally, we present the main theorem for using the imputed values as the dependent variable. Under standard classical assumptions, the theorem shows that OLS estimation with the imputed dependent variable remains unbiased.

Consider the relationship between out-of-pocket spending and total medical spending:[2]

$$p_i = D_i\beta + \gamma h_i + \epsilon_i, \qquad i \in \{M, H\}. \tag{1}$$

Here p denotes out-of-pocket spending, D is a vector of demographic and other controls, h is total medical spending, and ϵ is an error term. The index $i \in \{M, H\}$ denotes the donor (MEPS) and main (HRS) samples, respectively. Rearranging (1) yields the imputed total spending:

$$\hat{h}_i = \frac{p_i - D_i\hat{\beta}}{\hat{\gamma}}, \tag{2}$$

where $\hat{\beta}$ and $\hat{\gamma}$ are estimated in MEPS and $\hat{\gamma} \neq 0$.

Note that our imputation target, h_i, is an independent variable when we estimate γ and β in the donor dataset. This is an important consideration, since defining $\hat{h}_i$ as the dependent variable would cause estimation bias in the main sample.

[1] The appendix discusses alternative imputation approaches relevant to this study.
[2] See the appendix for a complete discussion of alternative imputation methods.

To study the large-sample properties of imputed health spending, it is useful to express imputed total medical spending, h_i, in terms of the observed variables. Substituting (1) into (2) yields the relationship between the true and imputed values:

$$\hat{h}_i = D_i \frac{\left(\beta - \hat{\beta}\right)}{\hat{\gamma}} + \frac{\gamma}{\hat{\gamma}} h_i + \frac{\epsilon_i}{\hat{\gamma}} \tag{3}$$

The imputation embeds both measurement error and systematic drift. As a result, using the imputed values in subsequent regressions can introduce bias, and even simple comparisons of means and variances may be misleading. To see this more clearly, define:

$$\bar{x} = \frac{\mathbf{1}_n' x}{n}, \qquad s_x^2 = \frac{(x-\bar{x})'(x-\bar{x})}{n}, \qquad s_{xy} = \frac{(x-\bar{x})'(y-\bar{y})}{n},$$

where $\mathbf{1}_n$ is an $n \times 1$ vector of ones. Let $\bar{x}$ denote the sample mean of the random variable x, let s_x^2 denote the sample variance of the random variable x, and let s_{xy} denote the sample covariance between the random variables x and y.

It is straightforward to show that if $\hat{\beta}$ and $\hat{\gamma}$ are inconsistent, then the imputed variable will be mean- and variance-inconsistent. The following assumptions ensure consistency of $\hat{\beta}$ and $\hat{\gamma}$:

Assumption 2.1. *The donor (MEPS) and main (HRS) samples are drawn from the same underlying population with respect to medical spending. Moreover,*

$$\mathbb{E}[\epsilon] = 0, \qquad \mathbb{E}[D'\epsilon] = 0, \qquad \mathbb{E}[h\,\epsilon] = 0.$$

A common concern in survey-to-survey imputation is measurement error. If the donor sample is measured with error, the consistency of $\hat{\gamma}$ and $\hat{\beta}$ is no longer guaranteed, so a simple OLS fit in the donor data need not be consistent. In our setting, however, the design of the MEPS survey limits the scope for substantial measurement error, making these assumptions reasonable.[3]

If the HRS and MEPS represent different underlying populations (so their true average medical spending differs), then a "consistent" imputation in the donor sense is not enough. At best, the imputed values in HRS will converge to the MEPS population mean, because the mapping is estimated to match MEPS. But the relevant benchmark for HRS is the HRS population mean. Therefore, even in large samples, the imputed

[3]The MEPS Medical Provider Component (MPC) reduces reporting error by collecting billing/record information directly from providers to supplement and verify household reports of charges and payments.

HRS mean can systematically differ from the true HRS mean—not because the imputation fails statistically, but because it is targeting the wrong population.

We do not face this issue because the target population is aligned across datasets: both MEPS and HRS are used to represent the U.S. elderly population. As a result, the donor-based relationship estimated in MEPS is intended to apply to the same underlying population observed in HRS, so the imputation targets the correct mean (and other moments) for HRS.

Therefore, under Assumption 2.1, the donor (MEPS) and main (HRS) samples share the same underlying population moments. Let μ and σ^2 denote the population mean and variance. Then Assumption 2.1 implies

$$\mu_{h_{\mathrm{M}}} = \mu_{h_{\mathrm{H}}} = \mu_h, \qquad \sigma^2_{h_{\mathrm{M}}} = \sigma^2_{h_{\mathrm{H}}} = \sigma^2_h.$$

Additionally, under Assumption 2.1, $(\hat{\beta}, \hat{\gamma})$ are consistent.[4] It follows that $p\text{lim } \bar{\hat{h}}_M = p\text{lim } \bar{h}_M$. The drift term vanishes as $\hat{\beta}$ converges to β, and the remaining error component converges to zero by the law of large numbers. Finally, the equality uses $\hat{\gamma}/\gamma \to 1$. Formally, Under assumption 2.1, equation (3) implies:

$$\begin{aligned} &p\text{lim } \bar{\hat{h}}_M = p\text{lim } \bar{h}_M = \mu_h, \\ &p\text{lim } s^2_{\hat{h}_M} = p\text{lim } s^2_{h_M} + \frac{1}{\gamma^2}\, p\text{lim } s^2_{\epsilon_M} = \sigma^2_h + \frac{1}{\gamma^2}\sigma^2_{\epsilon_M}. \end{aligned} \tag{4}$$

Within the sample, the imputed values $\hat{h}_{MEPS}$ are mean-consistent, i.e., they converge in probability to the true population mean μ_h. However, the sample variance of the imputed values exhibits a persistent positive bias due to the added measurement/imputation noise[5].

Additionally, properties of the imputed variable in the donor sample need not carry over to the out-of-sample imputations. Trivially, if the joint distribution of demographics and out-of-pocket spending is identical across the two samples, then all properties established above would be inherited. We do not impose that assumption. Applying (2) to the MEPS (donor) and HRS (main) samples yields:

$$\hat{h}_H = \hat{h}_M + \frac{1}{\hat{\gamma}}\left[(p_H - p_M) - (D_H - D_M)\,\hat{\beta}\right]. \tag{5}$$

[4]See the appendix for conditions and a proof.

[5]This asymptotic behavior follows directly from a classical measurement-error structure. The intuition is simple: we have two random variables x and y related by $y = x + u$, where u is white noise independent of x. It then follows that

$$\sigma^2_y = \sigma^2_x + \sigma^2_u \geq \sigma^2_x$$

Where the inequality is strict whenever $\sigma^2_u > 0$. In the MEPS imputation context, this classical errors-in-variables structure induces the inflation of the sample variance by the scaled noise term $\sigma^2_\epsilon/\gamma^2$, even though the mean remains unbiased in the limit.

Therefore, the average gap in imputed total spending across the two samples is driven by differences in out-of-pocket spending and differences in demographics (scaled by $\widehat{\gamma}$). In particular, if mean out-of-pocket spending is higher in HRS than in MEPS, then—holding demographic differences fixed—the imputed average total spending in HRS will exhibit a systematic upward bias.

Additionally, the standard deviation would still be biased as it was in the original case. The size of the bias depends on the distribution of the out-of-pocket spending and demographics in the two samples. Formally:[6]

$$\begin{aligned} \text{plim}\, \bar{\hat{h}}_H &= \mu_h + \tfrac{1}{\gamma}\, \text{plim}\left(\bar{p}_H - \bar{p}_M\right) - \tfrac{1}{\gamma}\, \text{plim}\left(\bar{D}_H - \bar{D}_M\right) \\ \text{plim}\, s^2_{\hat{h}_H} &= \sigma^2_h + \tfrac{1}{\gamma^2}\sigma^2_{\epsilon_M} + \tfrac{1}{\gamma^2}\, \text{plim}\left[s^2_{\left(p_H - D_H\hat{\beta}\right)} - s^2_{\left(p_M - D_M\hat{\beta}\right)}\right] \end{aligned} \tag{6}$$

Where $\bar{D}_i = \frac{\mathbf{1}'_n \times D_i\beta}{n}$ for $i \in \{H, M\}$.

Trivially, if out-of-pocket spending in the underlying populations of MEPS and HRS is identical, then the imputed mean of total health spending in the HRS sample is consistent.[7]
However, the standard deviation remains biased, at best by a constant positive drift.[8]
As shown in the data section, the demographic composition of MEPS and HRS is very similar for the selected age group. Hence, only an adjustment for differences in out-of-pocket spending is required, as formalized in the following theorem.

Theorem 2.1. *Define:*

$$\hat{h}_a \;=\; \hat{h}_H \;-\; \frac{1}{\hat{\gamma}}\left(\bar{p}_H - \bar{p}_M\right).$$

If the underlying demographic distributions in MEPS and HRS coincide and 2.1 holds, then $\hat{h}_a$ is mean consistent:

$$\text{plim}\, \bar{\hat{h}}_a \;=\; \mu_h,$$

[6]The first component is obtained by applying the mean operator to both sides of (5), then taking probability limits and using $\text{plim}\, \bar{\hat{h}}_M = \mu_h$.

For the second component, note that $s^2_{\hat{h}_H} = \frac{s^2_{p_H - D_H\hat{\beta}}}{\hat{\gamma}^2}$ and $s^2_{\hat{h}_M} = \frac{s^2_{p_M - D_M\hat{\beta}}}{\hat{\gamma}^2}$. Taking the difference yields

$$s^2_{\hat{h}_H} - s^2_{\hat{h}_M} = \frac{s^2_{p_H - D_H\hat{\beta}}}{\hat{\gamma}^2} - \frac{s^2_{p_M - D_M\hat{\beta}}}{\hat{\gamma}^2}.$$

Rearrange by moving $s^2_{\hat{h}_M}$ to the right-hand side and apply the $plim$ operator to both sides. Using (4) delivers the second component.

[7]Set $\mu_{p,HRS} = \mu_{p,MEPS} = \mu_p$. Then

$$\text{plim}\left(\bar{p}_H - \bar{p}_M\right) = \mu_{p_H} - \mu_{p_M} = \mu_p - \mu_p = 0.$$

Applying the same argument to the demographic variables yields $plim\, \bar{\hat{h}}_H = \mu_h$.

[8]That is,

$$\text{plim}\, s^2_{\hat{h}_H} = \sigma^2_h + \frac{1}{\gamma^2}\sigma^2_\epsilon.$$

Moreover, the variance term for $\hat{h}_H$ converges to the population variance up to a constant drift:

$$\text{plim}\, s^2_{\hat{h}_H} \;=\; \sigma^2_h \;+\; \frac{1}{\gamma^2}\sigma^2_{\epsilon_H}.$$

Proof. Starting from (6), the last term on the right-hand side vanishes. Hence,

$$\text{plim}\,\bar{\hat{h}}_H = \mu_h + \frac{1}{\gamma}\,\text{plim}(\bar{p}_H - \bar{p}_M)\,.$$

It follows that

$$\text{plim}\;\bar{\hat{h}}_a = \text{plim}\,\bar{\hat{h}}_H - \frac{1}{\gamma}\,\text{plim}\left(\bar{p}_H - \bar{p}_M\right) = \mu_h,$$

which establishes mean consistency of $\bar{\hat{h}}_a$.

For the variance, use (1) to write

$$p_i - D_i\beta = \gamma h_i + \epsilon_i.$$

Under mean independence, $\text{plim}\, s_{h_i,\epsilon_i} = 0$. Therefore,

$$\begin{aligned}\text{plim}\left[s^2_{(p_H - D_H\hat{\beta})} - s^2_{(p_M - D_M\hat{\beta})}\right] &= \text{plim}\left(\hat{\gamma}^2\,\left(s^2_{h_H} - s^2_{h_M}\right) + s^2_{\epsilon_H} - s^2_{\epsilon_M}\right)\\ &= \sigma^2_{\epsilon_H} - \sigma^2_{\epsilon_M},\end{aligned}$$

where the last equality follows from Assumption 2.1.[9] Substituting this expression back into (6) yields the stated variance result. ☐

Theorem 2.1 presents the consistency result. When working with imputed values, fixed and specific correction terms must be incorporated in the analysis of means and variances. We then examine the conditions under which imputed values can be used in other regression settings and establish the consistency of OLS.

[9] In particular, plim $s^2_{h_H}$ = plim $s^2_{h_M} = \sigma^2_h$.

3 Incorporating Imputed Variables into OLS Estimation

To illustrate the issue with using imputed values, consider a simplified single-variable version of (1), given by

$$p_i = \beta_0 + \beta_1 h_i + \epsilon_i \tag{7}$$

Where $i \in \{H, M\}$. Therefore imputed variables can simply be defined as:

$$\hat{h}_i = \frac{p_i - \hat{\beta}_0}{\hat{\beta}_1}$$

Therefore, the imputed values can be expressed as a function of the actual total health spending as follows:

$$\hat{h}_i = \frac{\beta_0 - \hat{\beta}_0}{\hat{\beta}_1} + \frac{\beta_1}{\hat{\beta}_1} h_i + \frac{\epsilon_i}{\hat{\beta}_1}.$$

Now suppose we seek to analyze the association between total health spending and wealth by estimating a simple linear regression model. Specifically, we relate individual health expenditures to wealth as follows:

$$h_i = \theta_0 + \theta_1 W_i + e_i, \tag{8}$$

where h_i denotes total health spending for individual i, W_i represents their level of wealth, θ_0 is the intercept, θ_1 measures the marginal effect of wealth on health spending, and e_i captures unobserved factors affecting health expenditures.

In this specification, total health spending is assumed to affect out-of-pocket spending exogenously, that is, $\sigma_{h_i,\epsilon_i} = 0$. Similarly, for consistency of the OLS estimates, we require $\sigma_{W_i,e_i} = 0$. Under this assumption, the slope coefficient is:

$$\theta_1 = \frac{\sigma_{h_i,w_i}}{\sigma^2_{w_i}}$$

Since we use imputed values in practice, we are estimating $p\lim \hat{\theta} = p\lim \frac{s_{\hat{h}_i,w_i}}{s^2_{w_i}}$. Without imputation, this would converge to θ by the law of large numbers. With imputed values, however, assuming $\hat{\beta}$ is consistent, we instead obtain:

$$p\lim \hat{\theta} = \frac{\sigma_{h_i,w_i}}{\sigma^2_{w_i}} + \frac{1}{\beta} \cdot \frac{\sigma_{\epsilon_i,w}}{\sigma^2_{w_i}} = \theta_1 + \frac{1}{\beta} \cdot \frac{\sigma_{\epsilon_i,w_i}}{\sigma^2_{w_i}}$$

The point of this exercise is straightforward. We have shown that using imputed values does not generally guarantee consistency of OLS estimates, which motivates the need to establish conditions under which consistency is preserved. In particular, some form of an exogeneity argument appears necessary when employing any imputation method.

More intuitively, in the ideal case, including every regressor that could plausibly affect total health spending through wealth in both the imputation equation (7) and the main regression (8) would resolve the issue, as this would satisfy the condition $\sigma_{\epsilon_i, w_i} = 0$. Formally:

$$\begin{aligned} p &= h\,\gamma + D\,\beta_1 + \epsilon \\ h &= w\,\alpha + D\,\beta_2 + u \end{aligned} \tag{9}$$

Where p, h, u and ϵ are $n \times 1$ vectors, D is $n \times k$, and β is $k \times 1$.

The first step is to isolate the imputation regression relationship for γ. To this end, define the projection matrix $P_D = D(D'D)^{-1}D'$ and the residual maker $M_D = I - P_D$. The residualized variable is then given by $\ddot{x} = M_D x$, which allows us to rewrite (9) as:

$$\begin{aligned} \ddot{p} &= \gamma \ddot{h} + \ddot{\epsilon} \\ \ddot{h} &= \alpha \ddot{w} + \ddot{u} \end{aligned} \tag{10}$$

The OLS estimator of α is consistent provided that residualized wealth is orthogonal to both residualized error terms, since:

$$p\text{lim } \hat{\alpha} = \alpha + p\text{lim}\,(\ddot{w}\ddot{w}')^{-1}\, p\text{lim}\,(\ddot{w}'\ddot{u}) + p\text{lim}\,(\ddot{w}\ddot{w}')^{-1}\, p\text{lim}\,(\ddot{w}'\ddot{\epsilon})$$

The key intuition is straightforward: valid imputation of total health spending requires controlling for all potential sources of confounding. In particular, any variable that affects total health spending through out-of-pocket expenditures must be included in the control set, otherwise the consistency argument would not hold.

We assume that, conditional on demographic controls D, wealth affects out-of-pocket spending only through total health expenditure — that is, $\mathbb{E}[\ddot{w}'\ddot{\epsilon}] = 0$. A potential violation is that wealthier households carry more generous insurance, so that w enters ϵ through the OOP share even at fixed h. This is an empirical concern rather than a failure of the identification strategy itself: the assumption is imposed at the population level, and demographic controls are intended to absorb systematic variation in the OOP share. Where the residual correlation remains a concern, a natural extension is to instrument for wealth using a variable uncorrelated with insurance generosity conditional on D; we leave this for future work.

A second caveat concerns inference. Crossley et al. (2022) show that the usual OLS standard errors from a regression with a BPP-imputed dependent variable understate the true asymptotic variance, and provide a correction. The point estimates reported in this paper remain consistent, but hypothesis tests based on uncorrected standard errors may over-reject. Applying the Crossley et al. (2022) correction is left for future work.

Under the additional assumption of exogeneity, $\mathbb{E}[\ddot{w}'\ddot{u}] = 0$, the remaining bias term vanishes, implying consistency of the imputation estimator. The following assumption formally summarizes this argument.

Assumption 3.1. *Both MEPS and HRS data sets are from the same underlying population, and* (9) *governs the relationship between total health spending, out of pocket health spending, and wealth. Further, in addition to 2.1:*

1. $E[h'\epsilon] = E[w'u] = 0$
2. $E[D'\epsilon] = E[D'u] = 0$

The exogeneity assumption on total health expenditure, $\mathbb{E}[h'e] = 0$, guarantees that wealth is uncorrelated with the residuals in the imputation equation, $\mathbb{E}[w'\epsilon] = 0$. This key condition is typically violated in survey data, as self-reported health spending is subject to measurement error — a concern that has been extensively documented in the literature. However, the structure of the MEPS dataset alleviates this concern, since health expenditure reports are verified directly by the health provider, lending credibility to the exogeneity assumption. The following Theorem and its proof summarize this section.

Theorem 3.1. *Consider applying the Ordinary Least Squares estimator to the following regression in the main dataset (HRS), where total health expenditure h is replaced by its imputed counterpart $\hat{h}$:*

$$\hat{h} = w\,\alpha + D\,\beta_2 + u$$

where $\hat{h}$ is imputed from a donor dataset (MEPS) via the regression equation:

$$p = h\,\gamma + D\,\beta_1 + \epsilon$$

and defined as in (2)*. Under Assumption 3.1, the OLS estimator $\hat{\alpha}$ is consistent.*

Proof. The first step is to show consistency of $\hat{\gamma}$. Define:

$$\hat{\gamma} = \left(\ddot{h}_i'\ddot{h}_i\right)^{-1}\ddot{h}_i'\ddot{p}_i$$

where $i \in \{M, H\}$. From (10), rewrite $\hat{\gamma}$ as:

$$\begin{aligned}
\hat{\gamma} &= \left(\ddot{h}_i'\ddot{h}_i\right)^{-1} \ddot{h}_i'\ddot{h}_i\,\gamma + \left(\ddot{h}_i'\ddot{h}_i\right)^{-1} \ddot{h}_i'\,\epsilon_i \\
&= \gamma + \left(\frac{\ddot{h}_i'\ddot{h}_i}{n_i}\right)^{-1} \frac{\ddot{h}_i'\,\epsilon_i}{n_i} \\
&= \gamma + \left(\frac{\ddot{h}_i'\ddot{h}_i}{n_i}\right)^{-1} \frac{h_i'\left(I - D_i(D_i'D_i)^{-1}D_i'\right)\epsilon_i}{n_i}
\end{aligned}$$

By the law of large numbers and Assumption 3.1, which implies both $\mathbb{E}[h'\epsilon] = 0$ and $\mathbb{E}[D'\epsilon] = 0$, we have that $\frac{\ddot{h}_i'\ddot{h}_i}{n_i}$ converges to a finite positive definite matrix and $\frac{\ddot{h}_i'\epsilon_i}{n_i} \to 0$, yielding consistency:

$$p\lim\,\hat{\gamma} = \gamma$$

The next step is to show consistency of $\hat{\beta}_1$. By definition:

$$\hat{\beta}_1 = (D_i'D_i)^{-1}D_i'(p_i - h_i\hat{\gamma})$$

Substituting $p_i = h_i\gamma + D_i\beta_1 + \epsilon_i$ from (9):

$$\hat{\beta}_1 = (D_i'D_i)^{-1}D_i'(h_i\gamma + D_i\beta_1 + \epsilon_i - h_i\hat{\gamma})$$

$$= \beta_1 + (D_i'D_i)^{-1}D_i'h_i(\gamma - \hat{\gamma}) + (D_i'D_i)^{-1}D_i'\epsilon_i$$

By the law of large numbers and Assumption 3.1, $\frac{D_i'\epsilon_i}{n_i} \to 0$. Combined with $p\lim\,\hat{\gamma} = \gamma$, the second and third terms vanish, yielding consistency:

$$p\lim\,\hat{\beta}_1 = \beta_1$$

Next, we characterize the imputed value $\hat{h}_i$ in terms of the true h_i. By definition:

$$\hat{h}_i = \frac{p_i - D_i\hat{\beta}_1}{\hat{\gamma}} = h_i\frac{\gamma}{\hat{\gamma}} + \frac{D_i(\beta_1 - \hat{\beta}_1)}{\hat{\gamma}} + \frac{\epsilon_i}{\hat{\gamma}}$$

Applying the residual maker:

$$\ddot{\hat{h}}_i = \ddot{h}_i\frac{\gamma}{\hat{\gamma}} + \frac{\ddot{\epsilon}_i}{\hat{\gamma}}$$

Finally, from (10) we define the OLS estimator of α as:

$$\hat{\alpha} = (\ddot{w}_i'\ddot{w}_i)^{-1}\,\ddot{w}_i'\ddot{\hat{h}}_i$$

Substituting $\ddot{\hat{h}}_i = \ddot{h}_i\frac{\gamma}{\hat{\gamma}} + \frac{\ddot{\epsilon}_i}{\hat{\gamma}}$:

$$\hat{\alpha} = \frac{\gamma}{\hat{\gamma}}(\ddot{w}_i'\ddot{w}_i)^{-1}\,\ddot{w}_i'\ddot{h}_i + \frac{1}{\hat{\gamma}}(\ddot{w}_i'\ddot{w}_i)^{-1}\,\ddot{w}_i'\ddot{\epsilon}_i$$

Substituting $\ddot{h}_i = \ddot{w}_i\alpha + \ddot{u}_i$:

$$\hat{\alpha} = \frac{\gamma}{\hat{\gamma}}\alpha + \frac{\gamma}{\hat{\gamma}}(\ddot{w}_i'\ddot{w}_i)^{-1}\,\ddot{w}_i'\ddot{u}_i + \frac{1}{\hat{\gamma}}(\ddot{w}_i'\ddot{w}_i)^{-1}\,\ddot{w}_i'\ddot{\epsilon}_i$$

Expanding using the residual maker:

$$\frac{\ddot{w}_i'\ddot{u}_i}{n_i} = \frac{w_i'\left(I - D_i(D_i'D_i)^{-1}D_i'\right)u_i}{n_i} = \frac{w_i'u_i}{n_i} - \frac{w_i'D_i(D_i'D_i)^{-1}D_i'u_i}{n_i}$$

$$\frac{\ddot{w}_i'\ddot{\epsilon}_i}{n_i} = \frac{w_i'\left(I - D_i(D_i'D_i)^{-1}D_i'\right)\epsilon_i}{n_i} = \frac{w_i'\epsilon_i}{n_i} - \frac{w_i'D_i(D_i'D_i)^{-1}D_i'\epsilon_i}{n_i}$$

By the law of large numbers, both expressions converge to zero under Assumption 3.1, which implies $\mathbb{E}[w'u] = 0$, $\mathbb{E}[D'u] = 0$, $\mathbb{E}[D'\epsilon] = 0$, and $\mathbb{E}[w'\epsilon] = 0$.[10] Combined with $p\text{lim}\ \hat{\gamma} = \gamma$, yielding:

$$p\text{lim}\ \hat{\alpha} = \alpha$$

□

4 Data and Sample Selection

Having established the consistency of the imputation estimator, we now describe the two datasets used to implement it empirically. The Medical Expenditure Panel Survey (MEPS) serves as the donor dataset from which we estimate the relationship between total and out-of-pocket health spending, while the Health and Retirement Study (HRS) is the main dataset in which we apply the imputation.

[10] Since $h = w\alpha + D\beta_2 + u$, the exogeneity assumption $\mathbb{E}[h'\epsilon] = 0$ implies $\alpha\mathbb{E}[w'\epsilon] + \beta_2\mathbb{E}[D'\epsilon] + \mathbb{E}[u'\epsilon] = 0$, which together with $\mathbb{E}[D'\epsilon] = 0$ and $\mathbb{E}[u'\epsilon] = 0$ yields $\mathbb{E}[w'\epsilon] = 0$.

Medical Expenditure Panel Survey (MEPS)

Since 1996, the Medical Expenditure Panel Survey has conducted large-scale surveys of individuals, families, and their healthcare providers, including doctors, hospitals, and pharmacies. MEPS collects data on the specific health services that Americans use, how frequently they use them, how much they cost, and how they are paid for. It currently consists of two major components: the Household Component and the Insurance Component. Data from households and their members in the Household Component are complemented by data from their medical providers, distinguishing MEPS from representative surveys that rely solely on self-reported expenditures. As discussed in the previous section, this provider-verified structure alleviates concerns about measurement error bias, which is the key property that motivates our use of MEPS as the donor dataset.

In this study, we use the Household Component of MEPS. A nationally representative subsample of households that participated in the National Health Interview Survey is used to collect data from families and individuals in selected communities across the United States. MEPS collects detailed information about each household member, including demographic characteristics such as sex, race, marital status, and education, as well as information on both total and out-of-pocket health spending. Through an overlapping panel design, data are collected for two calendar years for each panel of sample households, with a total of five rounds of interviews conducted over a two-and-a-half-year period. To be consistent with HRS sampling, we use MEPS Longitudinal Data Files, which follow respondents across the two years of each panel.

Health and Retirement Survey (HRS)

The Health and Retirement Study is a longitudinal study funded by the National Institute on Aging and the Social Security Administration that surveys a representative sample of approximately 20,000 Americans. Its in-depth interviews provide a growing library of multidisciplinary data well suited for research on the economic and health challenges of aging. In this study, we use the RAND HRS Longitudinal File, a cleaned and harmonized version of the HRS that includes derived variables covering demographics such as sex, race, marital status, and education, as well as imputations for out-of-pocket medical expenditures developed at RAND — the latter of which we replace with our own imputation from MEPS.

The HRS follows several cohorts, each interviewed in an initial wave and subsequently every two years. To keep the sample representative of the aging population, a new cohort is added every three waves. Our analysis pools across cohorts, retaining observations that satisfy our sample selection criteria described below.

Table 1: Comparing Demographic Means

HRS

	2006	2008	2010	2012	2014	2016
Sex	0.45	0.46	0.46	0.46	0.46	0.46
Race	0.92	0.91	0.91	0.91	0.92	0.91
Married	0.72	0.71	0.71	0.70	0.70	0.71
University Degree	0.28	0.29	0.32	0.33	0.34	0.35
Highschool Degree	0.87	0.88	0.90	0.91	0.92	0.93
Age	63.84	65.06	62.82	63.92	65.24	63.88

MEPS

	2006	2008	2010	2012	2014	2016
Sex	0.46	0.47	0.47	0.47	0.47	0.47
Race	0.91	0.90	0.90	0.89	0.89	0.89
Married	0.66	0.64	0.64	0.63	0.61	0.64
University Degree	0.34	0.36	0.39	0.40	0.42	0.41
Highschool Degree	0.86	0.86	0.88	0.88	0.85	0.89
Age	63.46	63.49	63.32	63.43	63.62	64.20

This table compares main regressors in the imputation process. Sex takes the value 1 if the respondent is male, and zero otherwise. Race takes the value of 1 if the respondent is white. Married represents the share of married couples and the last two variables illustrate the share of respondents with a university and high school degree.

Sample Selection

The first major distinction between the two surveys is that HRS records long-term health expenditures. To mitigate this discrepancy, we restrict the sample to individuals who have resided in nursing homes for one night or fewer, removing 20,629 observations (15.5 percent of the initial sample). Second, since out-of-pocket spending in HRS is imputed by RAND, there are notable outliers that may bias results. Following the RAND HRS documentation, we exclude observations with more than $20,000 in out-of-pocket health spending, removing 2,327 observations (1.7 percent). Third, since all individuals above age 85 are top-coded as 85 in MEPS, we restrict the sample to individuals aged between 51 and 84, excluding 7,144 observations (5.4 percent). Fourth, since we apply a log transformation to health spending variables, individuals with zero total or out-of-pocket spending are excluded, removing 12,695 observations (9.5 percent). These zeros are disproportionately concentrated in HRS, consistent with differences in how spending is recorded across the two surveys. Finally, due to differences in race definitions and oversampling practices across the two datasets, we focus exclusively on individuals identified as Black or White, dropping 6,865 observations (5.1 percent). After all restrictions, the analysis sample contains 83,693 observations. The analysis covers the period 2006–2016, corresponding to the waves for which both datasets are jointly available and comparable.

5 Imputation Consistency—Mean & Variance

Table 1 compares demographic means across the two datasets for the period 2006–2016. Given that HRS waves and longitudinal MEPS panels each span two years, we analyze the data in corresponding two-year periods.

The gender and racial compositions of the two samples are closely aligned throughout the period. Sex is virtually identical across all waves, and the share of White respondents follows a similar pattern in both datasets, ranging between 89 and 92 percent.

Differences are more pronounced for marital status and education. HRS respondents are approximately 6–9 percentage points more likely to be married than MEPS respondents, and MEPS respondents are 5–8 percentage points more likely to hold a university degree. Average age is comparable across the two datasets, though HRS exhibits more variation across waves—ranging from approximately 63 to 65, while MEPS remains more stable around 63–64. To formally assess whether these differences are statistically significant, we estimate the following regression separately for each wave:

$$D_i = \alpha + \beta\, Meps_i + \epsilon_i$$

where D_i represents each demographic variable and $Meps_i$ is an indicator for belonging to the MEPS dataset. Results are reported in Table 2.

Table 2: Differences in MEPS and HRS Demographics

	2006	2008	2010	2012	2014	2016
Male	0.01	0.01	0.00	0.01	0.01	0.01
	(0.31)	(0.43)	(0.68)	(0.30)	(0.24)	(0.55)
White	-0.01	-0.01	-0.01	-0.01**	-0.02***	-0.02***
	(0.18)	(0.07)	(0.07)	(0.01)	(0.00)	(0.00)
Married	-0.06***	-0.07***	-0.07***	-0.07***	-0.09***	-0.07***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
University Degree	0.06***	0.07***	0.07***	0.07***	0.08***	0.06***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Highschool Degree	-0.02**	-0.02***	-0.02***	-0.03***	-0.07***	-0.05***
	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Age	-0.37*	-1.57***	0.51***	-0.50**	-1.62***	0.32
	(0.07)	(0.00)	(0.01)	(0.01)	(0.00)	(0.11)
OOP	-94.35	-108.80*	-523.55***	-603.19***	-503.77***	-601.95***
	(0.11)	(0.08)	(0.00)	(0.00)	(0.00)	(0.00)

This table reports the estimated $\hat{\beta}$ *from regressing each demographic variable on a MEPS indicator, estimated separately for each wave. Numbers in parentheses are p-values. Stars follow the convention:* $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

As expected, differences in sex are statistically insignificant across all waves. Differences in race are small in magnitude (1–2 percentage points), though statistically significant in some later waves. Differences in marital status and education are statistically significant, and age differences reflect the cohort structure of the HRS and vary in sign across waves. In addition, differences in out-of-pocket spending are statistically significant and grow over time. Overall, these patterns point to the necessity of the correction proposed in equation (6).

Imputation Consistency Within MEPS (donor)

Table 3 compares the true and imputed total expenditure in log terms.[11]. The imputed values successfully reproduce the mean values, with differences of less than 0.1 in all years. This aligns with the theoretical prediction that, with consistent estimators, the imputation should closely replicate the mean.[12]

Table 3: True vs. Imputed Total Expenditure — MEPS

	Data				Imputation			
	mean	sd	iqr	skewness	mean	sd	iqr	skewness
2006	8.6	1.4	1.6	-0.10	8.6	1.7	2.0	-0.19
2008	8.7	1.4	1.7	-0.13	8.8	1.8	2.1	-0.20
2010	8.7	1.5	1.9	-0.11	8.7	2.0	2.4	-0.16
2012	8.6	1.6	1.9	-0.12	8.7	2.0	2.5	-0.19
2014	8.7	1.5	1.9	-0.13	8.8	2.1	2.5	-0.22
2016	8.8	1.6	1.9	-0.11	8.9	2.3	2.8	-0.17

This table summarizes different distribution moments in actual and imputed total medical expenditure. The left box represents actual moments and the right box represents the imputed data moments. In each box, the first column represents mean, the second represents standard deviation, the third represents interquantile range and the forth represents quantile based skewness.

In terms of standard deviation, the differences are more pronounced, and our imputation does not consistently reproduce the observed standard deviation. This outcome is consistent with equation (4), where we noted that even when estimates are consistent, the predicted standard deviation may differ due to the residual variance term $\sigma_\epsilon^2/\gamma^2$. Notably, the gap between the true and imputed standard deviations widens over time—from approximately 0.3 in 2006 to 0.7 in 2016—suggesting that the residual variance may be increasing across waves.

The widening gap between true and imputed standard deviations coincides with the entry of new cohorts into the HRS sample. The HRS employs a steady-state design, refreshing the sample every six years with younger birth cohorts aged 51–56 at enrollment. In 2010, the Mid Baby Boomers (born 1954–1959) entered the sample, and in 2016, the Late Baby Boomers (born 1960–1965) were added. These refreshment waves introduce respondents with systematically different health profiles and spending patterns than the incumbent sample.

Both interquartile ranges and standard deviations are smaller in the original data than in the imputed data,

[11]A linear specification ignores the concavity of the relationship between out-of-pocket and total spending: the out-of-pocket share of total spending declines with the level of expenditure, reflecting deductibles, coinsurance, and out-of-pocket maximums. The log transformation linearizes this relationship.

[12]Including an unrestricted constant in the log specification predicts that total health spending is less than out-of-pocket spending when the latter measure is relatively small. To avoid this issue, the demographic controls subsume the role of the intercept, allowing the level of out-of-pocket spending to vary across individuals through observed characteristics rather than an undifferentiated constant.

Table 4: Imputed Total Expenditure - MEPS vs. HRS

	MEPS				HRS				Mean Difference		SD Difference		Adjusted HRS
	mean	sd	iqr	skewness	mean	sd	iqr	skewness	Out of Pocket	Total	Out of Pocket	Total	mean
2006	8.64	1.72	2.00	-0.19	8.55	1.83	2.26	-0.16	-0.06	-0.09	-0.06	-0.09	8.63
2008	8.75	1.85	2.09	-0.20	8.79	1.79	2.24	-0.14	0.04	0.03	0.04	0.03	8.73
2010	8.74	2.02	2.44	-0.16	9.11	1.91	2.47	-0.10	0.27	0.36	0.27	0.36	8.70
2012	8.66	2.04	2.49	-0.19	9.10	1.93	2.47	-0.13	0.32	0.44	0.32	0.44	8.61
2014	8.78	2.11	2.55	-0.22	9.14	1.99	2.62	-0.10	0.29	0.35	0.29	0.35	8.70
2016	8.86	2.35	2.83	-0.17	9.34	2.20	2.93	-0.10	0.34	0.48	0.34	0.48	8.78

This table compares imputed total medical expenditure in MEPS and HRS. The first eight columns report the mean, standard deviation, interquartile range, and quantile-based skewness for each dataset. The next four columns decompose the mean and standard deviation differences into contributions from out-of-pocket spending. The final column reports the adjusted HRS mean following Theorem 2.1.

yet their ratio remains stable across all years and similar across the true and imputed distributions. In the true data, this ratio ranges from approximately 1.14 to 1.19, while in the imputed data it ranges from approximately 1.18 to 1.22, compared to 1.35 for a standard normal distribution. This indicates that our imputation method tends to preserve the relative spread of the actual data, with dispersion somewhat more concentrated than a normal distribution. Skewness levels are also comparable, although we rely on quantile-based skewness to mitigate the influence of outliers. The imputed data exhibits a slightly greater left skew in every wave.

Imputing Total Medical Spending in HRS

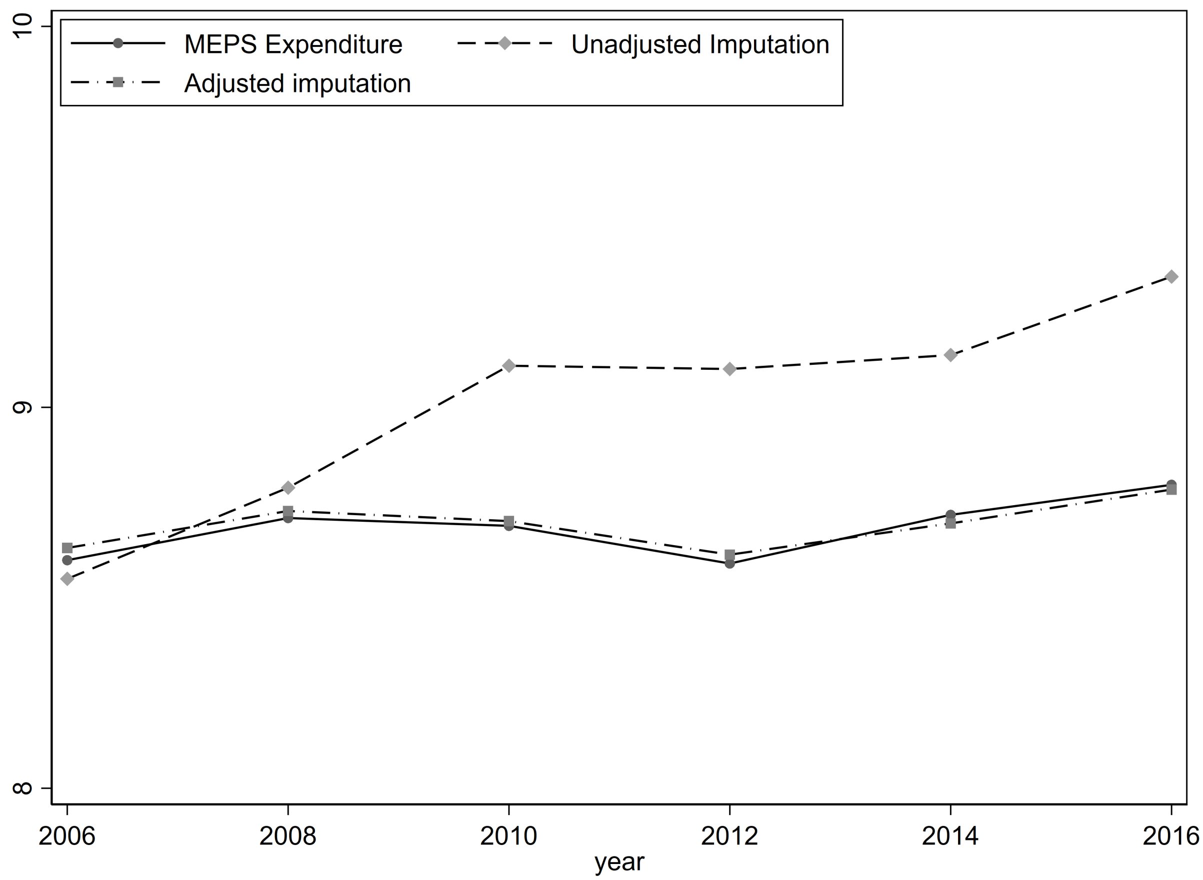
Table 4 compares the imputed total medical expenditure in MEPS and HRS. The mean values differ across the two datasets, with the gap widening over time. In 2006 and 2008, the difference in average imputed total spending is less than 10 percent. However, beginning in 2010, this gap increases substantially, reaching 30 to 48 percent by 2016.

Despite the differences in means, the dispersion of imputed spending is similar across the two datasets. The difference in standard deviation is less than 0.1 in all years, and the interquartile ranges follow a comparable pattern. This suggests that the imputed total spending in HRS preserves the spread observed in MEPS, consistent with the theoretical results in Section 2. The imputed HRS distribution exhibits slightly less left skewness than MEPS, though both distributions remain moderately left-skewed throughout the sample period.

Consistent with Equation (6), differences in out-of-pocket spending account for the bulk of the divergence in imputed means. As shown in Table 4, approximately 80 percent of the gap in mean imputed total spending is attributable to differences in out-of-pocket expenditures across the two samples. A similar pattern holds for the standard deviation.

The final column of Table 4 reports the adjusted HRS mean, computed following Theorem 2.1. After applying the out-of-pocket adjustment, the corrected HRS means closely track the imputed MEPS values across all waves. Figure 1 illustrates this alignment visually, confirming that the adjustment successfully removes the systematic bias induced by differences in out-of-pocket spending distributions.

Figure 1: Total Medical Spending Mean Comparison



Notes: This figure plots the mean of imputed total medical expenditure (in logs) across waves for three series: imputed MEPS, imputed HRS, and adjusted HRS. The adjusted HRS series applies the correction from Theorem 2.1, subtracting the scaled difference in mean out-of-pocket spending between the two datasets. After adjustment, the HRS means closely align with the MEPS values, confirming that differences in out-of-pocket spending distributions account for the divergence in unadjusted means.

6 Medical Spending–Wealth Gradient

Having established the validity of our imputation procedure, we now examine the relationship between total medical spending and wealth. A key advantage of imputing total medical expenditure in the HRS is access to detailed wealth measures unavailable in MEPS. While MEPS provides high-quality medical spending data verified by healthcare providers, it contains limited information on household assets. The HRS, by contrast, collects comprehensive data on wealth holdings, including housing, financial assets, pensions, and business equity. By combining imputed medical spending with the rich wealth data in the HRS, we can characterize the spending-wealth gradient across the wealth distribution.

Sample Construction

We analyze the relationship between medical spending and wealth separately for single-person and couple households. This separation is necessary because wealth in the HRS is measured at the household level, while medical spending is recorded for each individual. For single-person households, the individual and household unit coincide, so no aggregation is required. For couples, we must aggregate individual medical expenditures to the household level to match the unit of observation for wealth.

For couple households, we aggregate imputed medical spending as follows. Let $\hat{h}_1$ and $\hat{h}_2$ denote the imputed log medical expenditure for each spouse. Since log values cannot be summed directly, we first convert to levels, sum across spouses, and then take the logarithm:

$$\hat{h}_{hh} = \log\left(\exp(\hat{h}_1) + \exp(\hat{h}_2)\right) \tag{11}$$

This household-level measure, $\hat{h}_{hh}$, represents total medical spending for the couple and is directly comparable to household wealth.

We impose several sample restrictions. First, we limit the analysis to waves 8–13 (2006–2016), corresponding to the period for which imputed medical expenditure is available. Second, we retain only observations satisfying the imputation sample criteria: individuals aged 51–84, out-of-pocket spending at or below $20,000, positive total and out-of-pocket spending, and racial classification as Black or White. Third, we exclude individuals residing in nursing homes for more than one night. These restrictions ensure comparability between the HRS analysis sample and the MEPS donor sample used for imputation.

Additionally, we impose restrictions specific to each household type. For couples, we require that both spouses respond in each wave, that neither spouse remarries during the panel, and that the household does not split due to divorce or separation. These restrictions ensure that we observe the same couple unit throughout the panel. For singles, we exclude individuals who marry during the observation period.

Summary Statistics

Table 5 presents summary statistics for the analysis sample. The couple sample comprises 24,315 household-wave observations spanning 7,468 unique households, while the single sample contains 10,869 individual-wave observations from 4,379 unique individuals. All monetary values are expressed in real terms. For couples, demographic variables are measured at the household level: age is the average of both spouses, while education and race indicators capture whether either or both spouses possess the characteristic.

Table 5: Summary Statistics

	Couples		Singles	
	Mean	SD	Mean	SD
Medical Spending				
Log total medical expenditure	9.63	1.97	9.23	2.21
Log Out-of-pocket expenditure	8.12	1.17	6.98	1.36
Wealth				
Total wealth ($000s)	630.76	1,342.09	282.51	2,907.87
Demographics				
Age (average of spouses / individual)	65.21	9.16	64.30	9.23
Male (%)	—	—	26	44
Either spouse college (%) / College (%)	47	50	32	47
Both spouses college (%)	21	41	—	—
Either spouse White (%) / White (%)	87	34	65	48
Both spouses White (%)	81	39	—	—
Household-wave observations	24,315		10,869	
Unique households/individuals	7,468		4,379	

Notes: Sample restricted to waves 8–13 (2006–2016) with non-missing imputed medical expenditure. Wealth values in thousands of real dollars. For couples, the unit of observation is the household-wave; medical expenditure is aggregated across both spouses, and demographic variables are measured at the household level.

The comparison of out-of-pocket and total medical expenditure across household types is informative about the imputation procedure. Couples exhibit higher log medical spending than singles on both margins: log total medical expenditure averages 9.63 for couples versus 9.23 for singles, a gap of roughly 0.3 log points, while log out-of-pocket expenditure averages 8.12 for couples versus 6.98 for singles, a gap of 1.15 log points.

Total wealth is also different between these two groups. Couples' net worth averages to $630.76 thousand, while singles hold $282.51 on average.

The demographic composition of the two samples also differs. Singles are disproportionately female (74%) and less likely to be White (65% versus 81–87% for couples). Educational attainment differs somewhat: 47% of couple households have at least one college-educated spouse, though only 21% have both spouses with college degrees, compared to 32% of singles holding a degree. These compositional differences motivate our decision to analyze the two samples separately rather than pooling them with a household-type indicator.

Estimation

To characterize the medical spending–wealth gradient, we estimate the following pooled specification separately for couple and single households:

$$\hat{h}_{it} = \alpha + \sum_{d=2}^{10} \theta_d \cdot \mathbf{1}[W_{it} \in \text{decile } d] + D_{it}\beta + \delta_t + u_{it} \tag{12}$$

where $\hat{h}_{it}$ denotes imputed log total medical expenditure for household i in wave t, W_{it} is household net worth, D_{it} collects demographic controls, and δ_t are wave fixed effects.

The coefficients $\{\theta_d\}_{d=2}^{10}$ trace out the spending gradient across the wealth distribution, with the first (poorest) decile as the reference group. Standard errors are clustered at the household level to account for repeated observations on the same unit across waves.

Wealth deciles are constructed by pooling all household-wave observations within each sample and sorting by net worth. This approach maximizes statistical power relative to constructing deciles separately within each wave. Since all monetary values are CPI-adjusted, nominal price-level differences across waves do not affect the ranking. The wave fixed effects δ_t absorb any remaining aggregate shifts in medical spending—including those arising from policy changes such as the Affordable Care Act and compositional shifts from HRS refreshment cohorts—ensuring that the decile coefficients reflect cross-sectional variation in wealth rather than time-series trends.

For couples, the demographic controls include the average age of both spouses, an indicator for whether either spouse holds a college degree, and an indicator for whether either spouse is White. For singles, the controls are individual age, college degree, race, and gender. These variables correspond to the regressors used in the imputation equation estimated in Section 2.

Results

Table 6 reports the mean wealth, the estimated percentage difference in total medical spending, and the corresponding difference in out-of-pocket spending for each wealth decile, separately for couples and singles. By comparing the two spending measures estimated under the same specification, we can assess the extent to which out-of-pocket expenditure understates the medical spending–wealth gradient.

Table 6: Medical Spending by Wealth Decile: Total vs. Out-of-Pocket

	Couples			Singles		
Decile	Mean Wealth ($000s)	Total (%)	OOP (%)	Mean Wealth ($000s)	Total (%)	OOP (%)
1	−11.2	ref.	ref.	−26.5	ref.	ref.
2	40.9	27.1%	15.2%	0.1	−47.3%	−39.6%
3	91.5	27.1%	19.3%	4.0	−24.5%	−16.8%
4	151.6	32.2%	16.4%	18.2	1.3%	0.7%
5	226.8	27.0%	18.1%	46.2	5.1%	5.3%
6	327.3	45.2%	23.4%	85.2	27.0%	21.6%
7	472.1	45.3%	20.3%	141.6	12.5%	14.6%
8	681.5	42.6%	21.2%	237.7	7.6%	12.1%
9	1,084.9	58.1%	27.4%	432.9	27.4%	27.0%
10	3,243.0	82.5%	47.4%	1,887.3	37.4%	34.8%

Notes: Mean Wealth reports the average net worth within each decile in thousands of real dollars. Total and OOP columns report the percentage difference in spending relative to decile 1. Total refers to imputed total medical expenditure; OOP refers to observed out-of-pocket medical expenditure. Both regressions include wave fixed effects and demographic controls (age, education, race for both; gender for singles). Standard errors clustered at the household level.

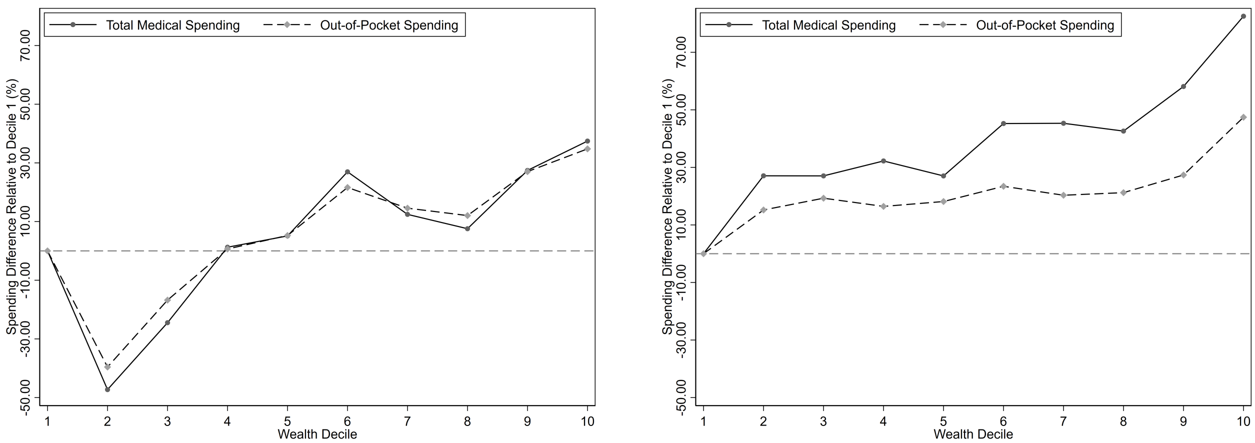
For singles, both measures exhibit a U-shaped pattern at the bottom of the wealth distribution, with deciles 2 and 3 spending less than the poorest decile. The two gradients track each other more closely than in the couple sample: at the tenth decile, total spending is 37 percent above the reference group compared to 35 percent for out-of-pocket, a gap of only 3 percentage points. Figure 2.a illustrates this pattern. The narrower gap for singles is consistent with less generous insurance coverage among single-person households, which would cause out-of-pocket spending to more closely approximate total expenditure.

For couples, the total spending gradient is rising from 27 percent at the second decile to 82 percent at the tenth decile relative to the poorest households. The out-of-pocket gradient is substantially flatter: it ranges from 15 to 27 percent across deciles 2 through 9 and rises to 47 percent only at the top decile. Figure 2.b displays both gradients. The gap between the two measures widens steadily across the wealth distribution, from approximately 12 percentage points at the second decile to 35 percentage points at the tenth.

Two findings emerge from Figure 2. First, out-of-pocket spending understates the medical spending–wealth

Figure 2: Total vs. Out-of-Pocket Spending by Wealth Decile

a. Singles b. Couples



Notes: Panel a. plots the estimated percentage difference in spending relative to decile 1 for single-person households, using both imputed total medical expenditure (solid) and observed out-of-pocket expenditure (dashed). Both regressions include wave fixed effects and demographic controls, with standard errors clustered at the individual level. Panel b. applies the same method to couple households.

gradient for couples: at the top decile, the total gradient (82.5%) is nearly twice as steep as the out-of-pocket gradient (47.4%), a gap of roughly 35 percentage points. Studies relying solely on out-of-pocket data would conclude that medical spending is only modestly related to wealth for this group, when in fact the total cost gradient is substantially steeper.

Second, the pattern differs sharply for singles. The total and out-of-pocket gradients track closely across the distribution, with a gap of only 2.6 percentage points at the top decile. Empirically, it provides direct support for the exclusion restriction underlying the imputation: if wealth affected out-of-pocket spending through the insurance share (conditional on total spending and demographics), we would expect to see a wedge between the two gradients. Among singles, this wedge is small, suggesting the residual correlation between wealth and the imputation error is limited in that sample.

7 Theoretical Framework

The decline in medical spending among low-wealth single households is substantial and warrants explanation. The U-shaped pattern in Figure 2 resembles the behavioral responses documented around eligibility thresholds in means-tested healthcare programs, in which households strategically adjust assets to qualify for coverage. To explore this conjecture, this section introduces a model in which individuals save to self-insure against health shocks in the absence of state-dependent securities. Households with assets just above the public-healthcare eligibility threshold face an incentive to decumulate, generating the U-shaped spending pattern documented above. We calibrate the model and compare its predictions to the empirical gradient.

Basic Model

Consider an environment where the pre-retirement single household is planning for a high-utilization state (e.g., nursing home). They know that they will move to this state according to the transition matrix Π:

$$\Pi = \begin{pmatrix} 1-\pi & \pi \\ 1 & 0 \end{pmatrix}$$

where $\pi \in (0,1)$ denotes the health shock probability, and the high-utilization state is absorbing and is reached with probability 1.

Let $\bar{a}$ denote the asset threshold for insurance eligibility, and let $\hat{m}$ denote the maximum amount covered by insurance when eligible. Out-of-pocket medical spending is denoted $p(.;\bar{a},\hat{m})$ and is defined as:

$$p(m,a;\bar{a},\hat{m}) = \begin{cases} 0, & \text{if } m \leq \hat{m} \text{ and } a < \bar{a} \\ m-\hat{m}, & \text{if } m > \hat{m} \text{ and } a < \bar{a} \\ m, & \text{if } a \geq \bar{a} \end{cases}$$

where m is total medical spending.

The separation of total and out-of-pocket spending is necessary to conceptualize moral hazard through a means-tested insurance system. The insurance system works as follows: it covers total spending if and only if assets are strictly below $\bar{a}$, and the coverage is up to the amount $\hat{m}$ — implying a co-pay of $(m-\hat{m})$. When ineligible, the household pays the full amount m out of pocket. Given how the insurance system operates, the moral hazard mechanism is activated through the incentive to fall into the free coverage zone — by decumulating assets. In the high-utilization state, there is one source of income — saving flows. The stay-in facility provides the consumption needs, depending on the household contribution m. The Bellman equation, therefore, is as follows:

$$
\begin{aligned}
V^h(a) &= \max_{a',\, m} \left\{ u\,(m) + \beta\, V^h(a') \right\} \\
\text{st. } & a' + p(m, a; \bar{a}, \hat{m}) = (1+r)a \\
& a' \geq \underline{a}, \quad m \geq 0
\end{aligned}
\tag{BEH}
$$

with $p(.; \bar{a}, \hat{m})$ defined as above.

It is worth noting that there is no precautionary saving motive in the high-utilization state by design. However, there is still an incentive to save — to smooth consumption: the more you save, the better you can pay. It would be interesting to add the facility problem to this — such a facility would provide different consumption levels in a non-continuous manner, reflecting the observation that different facilities provide different qualities.

The model deviates from the traditional set-up as follows: medical spending and consumption increase together in the high-utilization state. That is, an increase in medical spending does not crowd out the consumption expenditure, but actually increases it.

Additionally, in the simple framework, we assume $c(m) = m$. This does not need to be the case in general. One can consider a more sophisticated relationship between the two — and estimate an empirical relationship between consumption and medical expenditure in the high-utilization state. But that is not the focus of the simple framework.

The low-utilization state problem is the textbook consumption-saving problem, with health shocks arriving with probability π. Given their income, which does not fluctuate in the simple model, and their assets, households decide how much to consume and how much to save:

$$
\begin{aligned}
V^l(a) &= \max_{a',\, c} \left\{ u(c) + \beta\, \left[(1-\pi) V^l(a') + \pi V^h(a') \right] \right\} \\
\text{st. } & a' + c = (1+r)a + y \\
& a' \geq \underline{a}, \quad c \geq 0
\end{aligned}
\tag{BEL}
$$

The precautionary saving motive appears in the low-utilization state. The uncertainty about the future state of health and the lack of state-dependent securities drives the consumer to save. The problem, therefore, shrinks to the ratio of the marginal utilities gained from consumption in the two states, c/m — which is equal to $c/c(m)$ in a more general model.

8 Conclusion

This chapter developed an imputation procedure for estimating total medical expenditure in the Health and Retirement Study using the Medical Expenditure Panel Survey as a donor dataset. Building on the method of Blundell et al. (2004), we established conditions under which the imputed values are mean-consistent and showed that the sample variance exhibits a known positive bias from imputation noise. We further proved that using the imputed variable as the dependent variable in an OLS regression preserves consistency of the slope coefficients under standard exogeneity assumptions, which are supported by the provider-verified structure of the MEPS data.

The imputation procedure closely reproduces the mean of total medical expenditure, as documented in Figure 1. After adjustment, the imputed total health spending in HRS tracks the actual total health spending in MEPS closely across all six waves, validating the theoretical prediction that the imputation targets the correct population moments once cross-survey differences in out-of-pocket spending are accounted for.

A second implication of the theoretical framework is that imputed total medical expenditure can be used as a dependent variable. The decile gradient results in Table 6 are consistent with this claim. For single households, the two spending measures generate similar gradients, while for couples, out-of-pocket spending understates the spending–wealth gradient. Wealthier couples incur substantially higher total medical expenditures, but their out-of-pocket exposure is compressed by more generous coverage, attenuating the gradient that researchers would observe using only self-reported cost-sharing data. The imputed measure recovers this hidden spending, confirming that reliance on out-of-pocket data alone leads to a systematic understatement of the medical expenditure–wealth relationship among couple households.

Finally, our results are consistent with Blundell et al. (2004) in a different setting. In a different context, using different datasets, they use the demand for food estimated in the Consumer Expenditure Survey to impute total nondurable consumption in the Panel Study of Income Dynamics. They find that, using adjustments for differences in food expenditure, one can use the imputation method to recover total nondurable spending with mean consistency.

References

Attanasio, O. and Pistaferri, L. (2014). Consumption Inequality over the Last Half Century: Some Evidence Using the New PSID Consumption Measure. *American Economic Review*, 104(5):122–126.

Blundell, R., Pistaferri, L., and Preston, I. (2008). Consumption inequality and partial insurance. *American*

Economic Review, 98(5):1887–1921.

Blundell, R. W., Pistaferri, L., and Preston, I. (2004). Imputing consumption in the psid using food demand estimates from the cex. Technical report, IFS Working Papers.

Bollinger, C. R. and Minier, J. (2015). On the Robustness of Coefficient Estimates to the Inclusion of Proxy Variables. *Journal of Econometric Methods*, 4(1):101–122.

Browning, M. and Crossley, T. (2009). Are two cheap, noisy measures better than one expensive, accurate one? *The American Economic Review*, 99(2):99–103.

Browning, M. and Leth-Petersen, S. (2003). Imputing consumption from income and wealth information. *The Economic Journal*, 113(488):F282–F301.

Campos, R. G. and Reggio, I. (2014). Measurement error in imputation procedures. *Economics Letters*, 122(2):197–202.

Crossley, T. F., Levell, P., and Poupakis, S. (2022). Regression with an imputed dependent variable. *Journal of Applied Econometrics*, 37(7):1277–1294.

Palumbo, M. G. (1999). Uncertain medical expenses and precautionary saving near the end of the life cycle. *The Review of Economic Studies*, 66(2):395–421.

Skinner, J. (1987). A superior measure of consumption from the panel study of income dynamics. *Economics Letters*, 23(2):213–216.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. MIT press.

Ziliak, J. P. (1998). Does the choice of consumption measure matter? an application to the permanent-income hypothesis. *Journal of Monetary Economics*, 41(1):201–216.